

STM8L 和STM8AL 入门

Preface

This application note supplements the information in the STM8L and STM8AL data sheets by describing the minimum hardware and software environment required to build an application around these 8-bit microcontroller devices.

A brief introduction to the main hardware components is provided. Detailed descriptions of the power supply, analog-to-digital converter (ADC), clock management, and reset control are given. In addition, some hardware suggestions are offered. This application note also includes detailed reference design schematics for the main components. The STM8 development tools and software toolchain are applicable to STM8L, STM8S, STM8AL, and STM8AF, and this is described in [Sections 8](#) and [9](#). [Section 10](#) describes how to set up the STM8 development environment. Finally, [Section 11](#) provides a list of related documentation and online support resources.

Table 1. Applicable Products

Product series	Product part number
microcontroller	-STM8L051/52
	-STM8L101
	-STM8L151C2/K2/G2/F2 and STM8L151C3/K3/G3/F3
	-STM8L151x4, STM8L151x6, STM8L152x4, STM8L152x6
	-STM8L151x8, STM8L152x8, STM8L151R6, STM8L152R6, STM8L162R8, STM8L162M8
	-STM8AL313x, STM8AL314x, STM8AL316x, STM8AL3L4x, STM8AL3L6x

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1 Summary of Hardware Requirements

To build applications around STM8L or STM8AL, the application board must at least provide the following features:

- Power supply (essential)
- Clock Management (Optional)

Reset management (optional)

- Debugging tool support: Single Wire Interface Module (SWIM) connector (optional)

2 Power Supply

2.1 Power Supply Overview

The external power supply range for STM8L or STM8AL can be from 1.65 V to 3.6 V (the power supply range for STM8L05xxx is 1.8 V to 3.6 V). For the medium-capacity STM8L15xxx, medium-capacity STM8AL31xx/STM8AL3Lxx, and high-capacity STM8L15xxx/STM8L162xx with BOR, the supply voltage must be above 1.8 V at power-on and can be reduced to 1.65 V during power-off.

An on-chip power management system provides a constant digital power supply to the core logic with both normal and low-power modes. This ensures that the logic circuits consume a constant current throughout the entire voltage range. It can also detect voltage drops and generate a reset to prevent abnormal behavior.

STM8L and STM8AL devices offer:

Depending on the package, there is a pair of (VDD/VSS) pads, or several pairs of pads ranging from 1.65 V or 1.8 V to 3.6 V (VDDx/VSSx).

All VDDx and VSSx pins must be at the same potential. Each VDDx pin must be connected to VDD through two external decoupling capacitors: a 100 nF ceramic capacitor and a 1 μ F tantalum or ceramic capacitor.

The STM8L15xxx, STM8AL31xx and STM8L162xx devices also provide in certain packages:

A pair of pads specifically designed for simulating the power supply of the analog function pins VDDA/VSSA. VDDA and VSSA must be at the same potential, just as VDD and VSS are. For [more detailed](#) information, refer to [Section 3: Analog-to-Digital Converter \(ADC\)](#). The VDDA pin must be connected to two external decoupling capacitors (one 100 nF ceramic capacitor + one 1 μ F tantalum or ceramic capacitor). Additional measures can be taken to filter analog noise: VDDA can be connected to VDD through a ferrite bead.

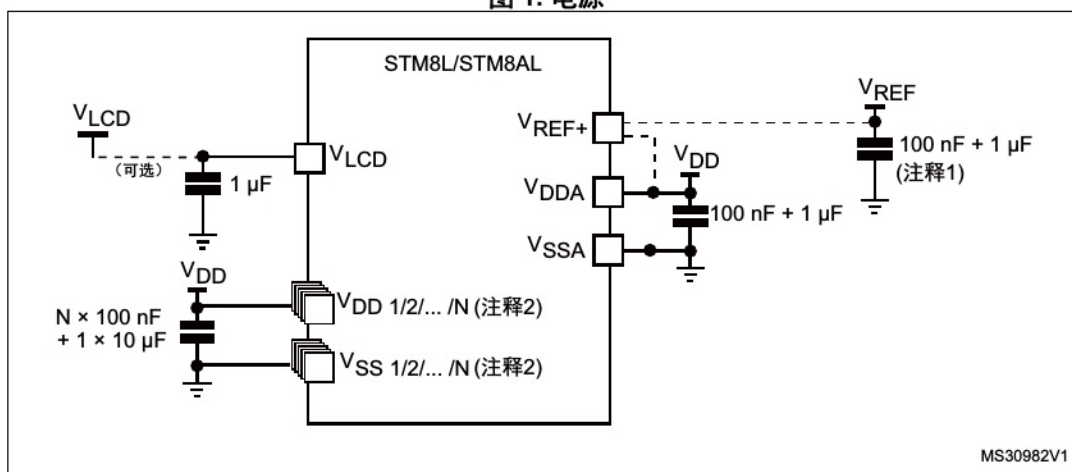
The STM8L152xx, STM8AL3Lxx and STM8L162xx devices manage the supply voltage required for the LCD in three different ways (see [Figure 1](#)):

If the LCD feature is not used, connect the VLCD pin to VDD.

Apply the voltage added to the LCD to VLCD.

3. Connect the VLCD pin to a 1 μ F capacitor to provide the correct voltage to the STM8L152xx/STM8L162xx/STM8AL3Lxx through the programmable LCD booster.

图 1. 电源



Optional: If a separate external reference voltage is applied to VREF+, a 100 nF and a 1 µF capacitor must be connected to this pin. VREF+ is connected to VDDA or VREF.

2. N is the number of VDD and VSS inputs.

Note: The capacitor must be placed as close as possible to the power connection of the device.

A crystal or resonator can be placed on *OSCIN/OSCOUT*. The resonator must be as close as possible to the connection of the *OSCIN* and *OSCOUT* pins. The load capacitor must be as close as possible to the *VSS* connection.

2.2 Main operating voltage

The STM8L and STM8AL devices are fabricated using a 0.13 µm process. The cores and I/O peripherals of the STM8L and STM8AL devices require different power supplies. In fact, the STM8L and STM8AL devices have an internal voltage regulator with a nominal target output of 1.8 V.

2.3 Power-on/Power-down Reset (POR/PDR)

The input power supply for the main regulator and the low-power regulator is monitored by the power-on/power-off reset circuit. The monitoring voltage starts from 0.7 V.

When the power is turned on, POR/PDR keeps the device in reset until the power supply voltages (VDD and VDDA) reach their specified operating regions. The maximum power must be observed for time $t_{VDD[max]}$, as the internal reset for power stabilization remains for approximately 1 ms.

At power-on, a predefined reset of less than 0.7 V should be maintained. The upper limit for reset release is defined in the electrical characteristics section of the product data sheet.

The delay (POR > PDR) is used to ensure accurate detection of voltage rise and fall.

When the power supply voltage drops below the VPDR threshold value (isolated and repetitive events), POR/PDR will also generate a reset.

When $t_{VDD[max]}$ is satisfied, correct device reset during power-on can be guaranteed.

When $V_{PDR} < V_{DD} < V_{DD[min]}$, it is recommended to use the internal BOR or external reset circuit (especially for

The STM8L101xx devices) should be ensured to reset correctly during power loss.

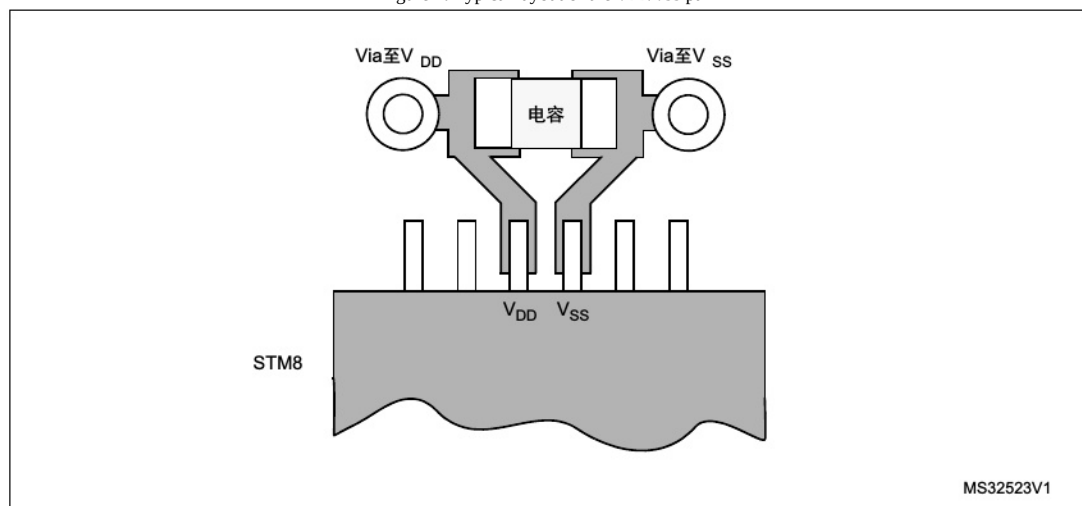
To achieve better power supply monitoring, the STM8L15xxx, STM8L162xx, STM8AL31xx and STM8AL3Lxx offer an under-voltage reset (BOR) and programmable voltage detector (PVD) for early detection of voltage drops.

Suggestion

All pins need to be correctly connected to the power supply. These connections, including pads, traces and vias, should have as low an impedance as possible. Typically, this can be achieved by using wide traces, and it is best to use a dedicated power supply layer in a multi-layer printed circuit board (PCB).

In addition, each power supply pair should be decoupled using a filter ceramic capacitor (100 nF) and an electrolytic capacitor (1 to 2 μ F), which are connected in parallel with the STM8L/STM8AL device. These ceramic capacitors should be placed on the opposite side of the PCB as close as possible to or below the appropriate pins. Their typical values range from 10 nF to 100 nF, but the exact value depends on the application requirements. Figure 2 shows a typical layout for such VDD/VSS pairs.

Figure 2. Typical layout of the VDD/VSS pair



3 Analog-to-Digital Converter (ADC)

This chapter does not apply to the STM8L101xx devices.

3.1 Analog Circuit Power Supply

For some packages, the ADC unit has an independent analog supply voltage that is isolated from the input pin V_{DDA} , thus providing a very clean power supply for the ADC. The analog voltage V_{DDA} should be the same as the digital supply voltage on the V_{DD} pin. To filter out certain noises, a ferrite bead can be placed between V_{DD} and V_{DDA} . The ferrite bead should be selected based on the frequency to be filtered.

Some packages also provide an independent external analog reference voltage input for the ADC unit on the V_{REF+} pin. This offers better accuracy for low-voltage inputs, as follows:

The V_{REF+} pin can be connected to the external power supply V_{DDA} . If a separate external reference voltage is applied to V_{REF+} , a 100 nF and a 1 μ F capacitor must be connected to this pin. To compensate for peak current consumption on V_{REF} , when the sampling rate is low, the 1 μ F capacitor can be increased to a maximum of 10 μ F. In all cases, V_{REF+} must be maintained between 2.4 V and V_{DDA} . If V_{DDA} is less than 2.4 V, V_{REF+} must equal V_{DDA} . In devices without an internal V_{REF+} pin, this input is internally connected to V_{DDA} .

- V_{REF-} (Input, Analog Reference Negative): The low-end/negative reference voltage is internally connected to V_{SSA} .

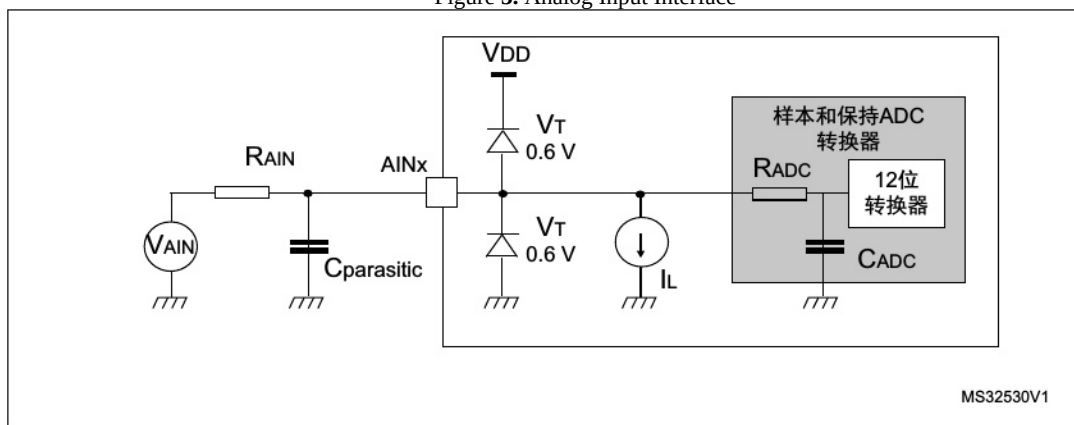
3.2 Simulated input

The device contains up to 28 analog input channels (including four fast channels), each channel multiplexing an I/O, and only one channel is converted by the ADC at a time.

The maximum value of the external input resistance (R_{AIN}) is 50 k Ω . If R_{AIN} is less than 0.5 k Ω , the four fast channels can perform conversions at the maximum speed of 1 MHz.

Please refer to **Figure 3**.

Figure 3. Analog Input Interface



For more detailed information, please refer to the application data manual and reference manual.

4 Clock Management

The STM8L101xx devices do not have an external clock, so no preventive measures are required.

4.1 Overview of Clock Management

The STM8L05xxx, STM8L15xxx, STM8L162xx, STM8AL31xx and STM8AL3Lxx devices offer a flexible way to select the core and peripheral (ADC, memory, digital peripherals) clocks. The devices feature both internal and external clock source inputs, each with a high-speed and low-speed version. Through a programmable prescaler, any of the four clocks can be used for the CPU and most peripherals. The I/O can be programmed as an output clock (CCO) to output one of the four clocks (with or without prescaling).

The signal leaving I/O represents an output clock (CCO) that is divided by the division factor.

4.2 Internal clock

The STM8L and STM8AL devices feature two types of internal clocks: a high-speed internal clock (HSI) with a frequency of 16 MHz and a low-speed internal clock (LSI) with a frequency of 38 kHz.

After reset, the CPU starts up at an internal RC (HSI clock signal) frequency divided by 8, that is, 2 MHz.

4.3 External clock

STM8L05xxx, STM8L15xxx, STM8L162xx, STM8AL31xx and STM8AL3Lxx feature two types of external clocks: a high-speed external clock (HSE) with a frequency up to 16 MHz and a low-speed external clock (LSE) with a frequency of 32.768 kHz.

4.3.1 HSE Clock

The STM8L05xxx, STM8L15xxx, STM8L162xx, STM8AL31xx and STM8AL3Lxx devices can be connected to an external crystal or an external oscillator.

When an external clock is not used, *OSCIN* and *OSCOU*T can be used as general-purpose I/Os.

Figure 4 shows the connection of the external clock.

External clock

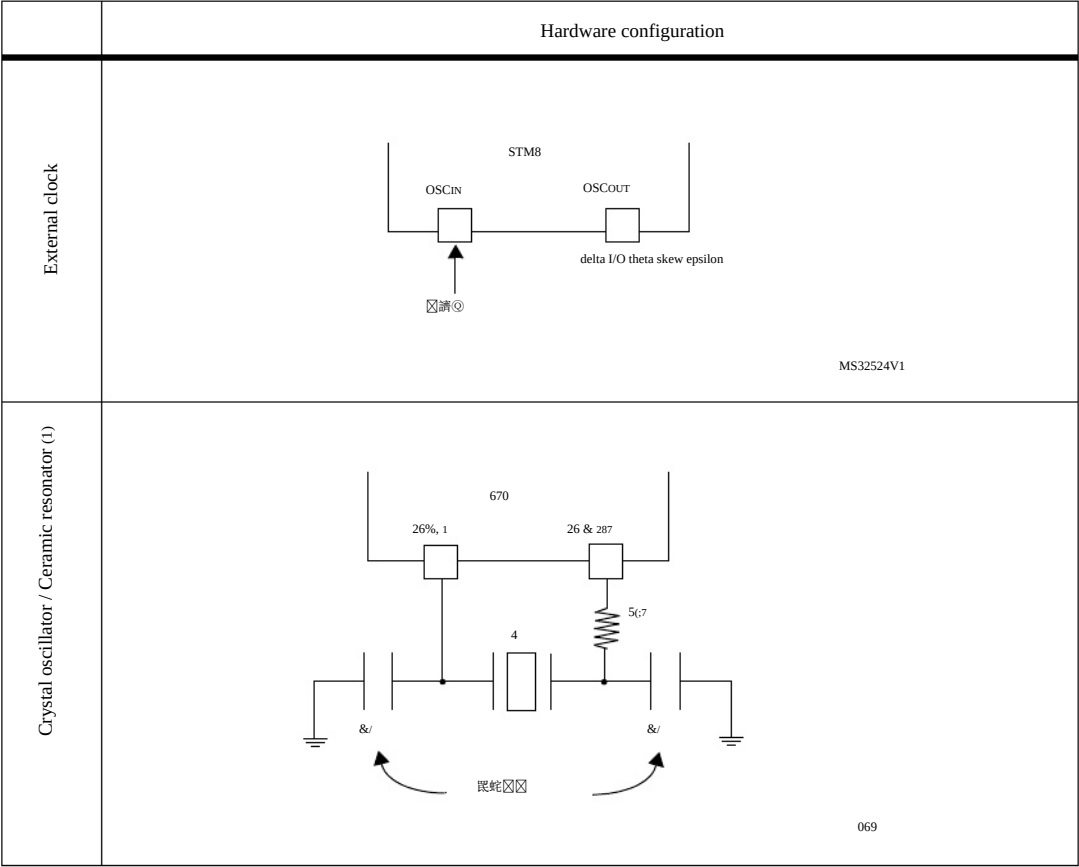
- Frequency: 0 kHz ... 16 MHz
- Input lag: 100 mV

Note: Without a prescaler, the duty cycle must be maintained within a maximum of 45/55% at high speeds.

Crystal oscillator / Ceramic resonator

- Frequency range: 1 to 16 MHz
- Stable time: Programmable from 1 to 4096 cycles
- Oscillator mode: Preferred fundamental frequency
- Output duty cycle: maximum 55/45%
- I/O: Standard I/O pins, multiplexed with OSCIN and OSCOUT
- Load: 10 to 20 pF
- Maximum drive level: at least 100 μ W

Figure 4. HSE Clock Source



The value of REXT depends on the crystal oscillator characteristics. A 0 Ω resistor is suitable for most oscillators, but it is not the optimal choice. The typical range is 5 to 6 RS (resonant series resistance). For fine-tuning the value of REXT, please refer to AN2867 (ST Microcontroller Oscillator Design Guide).

The values of the load capacitors CL1 and CL2 largely depend on the type and frequency of the crystal oscillator. Users can refer to the data sheet of the crystal oscillator manufacturer to select the capacitors. For the best oscillation stability, CL1 and CL2 usually have the same value. The typical range is from less than 20 pF to 40 pF (load: 10 to 20 pF). The parasitic capacitance of the board layout also needs to be considered, and a few pF should be added to the component values (refer to AN2867).

The clock security system prevents any fatal errors from CPU failure in HSE and safely switches to HSI.

Suggestion

On the PCB layout, all connections should be as short as possible. Any additional signals, especially those that may interfere with the oscillator, should be properly shielded to keep them locally isolated from the PCB area of the oscillator circuit.

4.3.2 LSE Clock

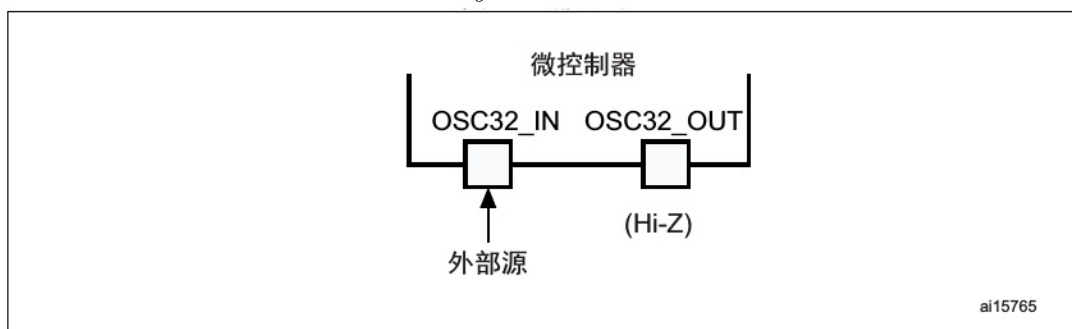
The low-speed external clock signal (LSE) has two clock sources:

- External crystal oscillator / ceramic resonator for LSE (see [Figure 6](#))
- LSE external clock for the user (see [Figure 6](#))

External source (LSE bypass)

In this mode, an external clock source must be provided. The frequency must be 32.768 kHz. The OSC_IN pin must be driven by an external clock signal with a duty cycle of approximately 50% (square wave, sine wave or triangular wave), while the OSC_OUT pin must remain in a high-impedance state (please refer to [Figures 5](#) and [6](#)).

Figure 5. External Clock



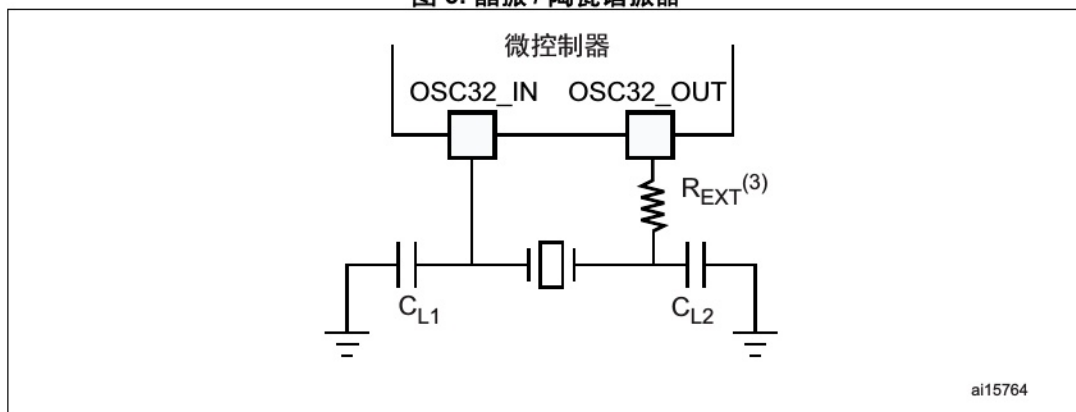
The OSC32_IN and OSC32_OUT pins can also be used as GPIO, but it is recommended not to use them as both RTC and GPIO pins in the same application.

External crystal / Ceramic resonator (LSE crystal)

The LSE crystal oscillator operates at 32.768 kHz and is a low-speed external crystal oscillator or ceramic resonator. It can be used as the clock source for the real-time clock (RTC) peripheral to provide clock/calendar or other timing functions, featuring low power consumption and high accuracy.

The resonator and load capacitor must be placed as close as possible to the oscillator pins to minimize output distortion and the start-up stabilization time. The value of the load capacitor must be adjusted appropriately depending on the selected oscillator.

图 6. 晶振 / 陶瓷谐振器



To avoid exceeding the maximum values of CL1 and CL2 (15 pF), it is strongly recommended to use a resonator with a load capacitance CL of 7 pF. For more detailed selection, please refer to the gmcrit calculation section in AN2867 (ST Microcontroller Oscillator Design Guide).

The OSC32_IN and OSC32_OUT pins can also be used as GPIO pins, but it is recommended not to use them as both RTC and GPIO pins in the same application.

The value of REXT depends on the crystal oscillator characteristics. A 0 Ω resistor is suitable for most oscillators. The typical range is 5 to 6 k Ω . For fine-tuning the value of REXT, please refer to AN2867 (ST Microcontroller Oscillator Design Guide).

5 Reset control

5.1 Overview of Reset Management

The reset pin is a 3.3 V bidirectional I/O. After startup, it can be programmed by software as a general-purpose I/O.

Its output buffer includes a $\sim 45\text{ k}$ pull-up resistor, with the drive capability locked at $I_{O1MIN} = 2\text{ mA @ } 0.45\text{ V}$ (in the 1.8V to 3.6V range). The output buffer is simplified to an n-channel MOSFET (NMOS). The receiver includes an interference filter, and here the output buffer has a $20\text{ }\mu\text{s}$ delay.

There are many reset sources, including:

External reset through the NRST pin

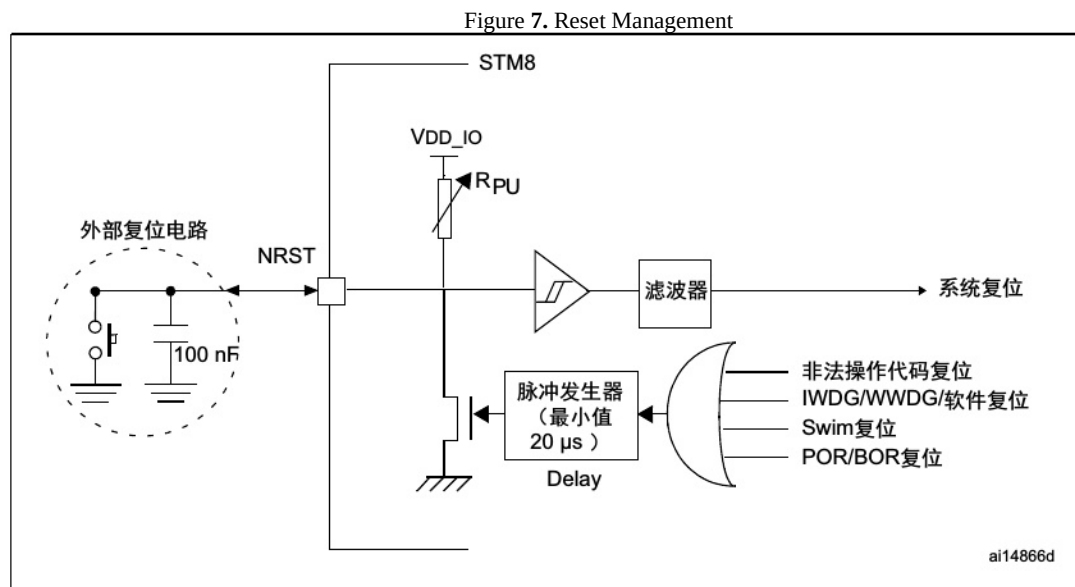
- Power-on reset (POR) and brownout reset (BOR): During power-on, POR keeps the device in reset until the supply voltages (V_{DD} and V_{DDX}) reach the voltage level at which BOR starts to operate. The STM8L101xx devices have only one POR.

Independent Watchdog Timer Reset (IWDG)

The Window Watchdog (WWDG) also features a software reset: applicable only to STM8L05xx, STM8L15xxx, STM8L162xx, STM8AL31xx and STM8AL3Lxx.

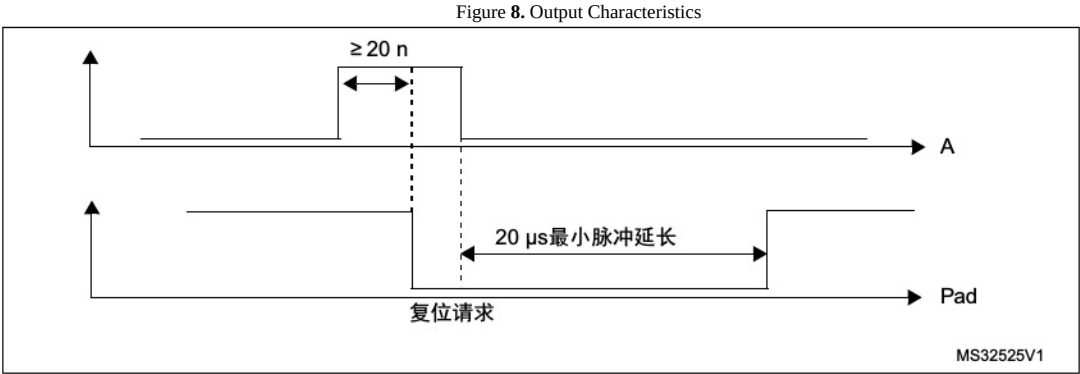
- SWIM reset: An external device connected to the SWIM interface can request the SWIM module to generate a reset of the microcontroller.
- Illegal operation code reset: If the executed code does not match any operation code or the previous value of the byte, a reset will be generated.

Figure 7 shows a simplified I/O reset diagram.



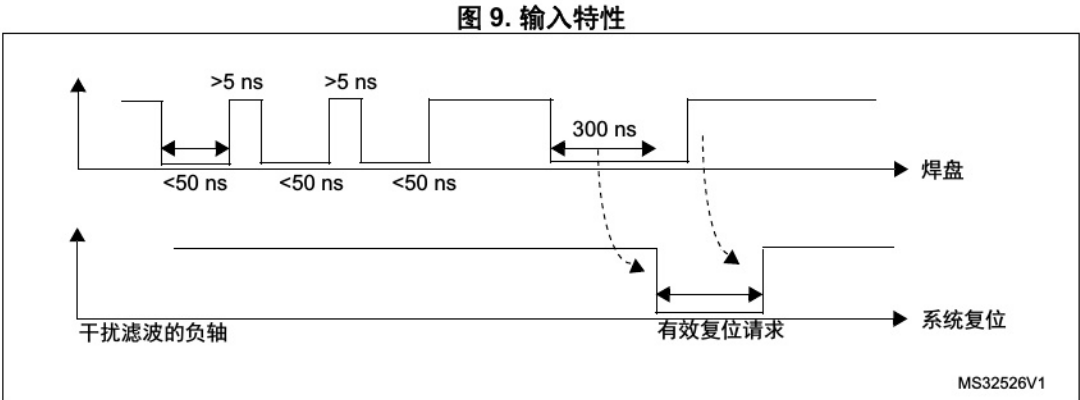
5.1.1 Output Characteristics

- A pulse duration of ≥ 20 ns on the internal output buffer ensures an effective pulse at the pin.
- After the valid pulse is identified, it can be guaranteed that there will be a pulse of at least $20\ \mu\text{s}$ on the pin starting from the falling edge of A.



5.1.2 Input Characteristics

- All pulses with a duration less than 50 ns are filtered out.
- All continuous/bursty spikes with a 1/10 ratio are filtered out. This means that when a 5 ns interval appears between the spikes (a ratio of 1/10), negative spikes lasting up to 50 ns are necessarily filtered out.
- All pulses with a duration greater than 300 ns are identified as valid pulses.



5.2 Hardware reset implementation

STM8L and STM8AL can be powered on correctly without an external reset circuit. It is recommended to use only one pull-down capacitor (see [Figure 7](#)). However, charging and discharging the pull-down capacitor through the internal resistor will have a negative impact on the device's power consumption. Therefore, the recommended value of this capacitor (100 nF) can be reduced to 10 nF to limit this power consumption.

After the POR value reaches (1.35 V to 1.65 V), the reset state of STM8L101xx is released for 1 ms. During this period, VDD should be within the range of 1.65 V to 3.6 V.

For medium-capacity devices operating above 1.8 V at power-on and medium-capacity and above as well as high-capacity devices: after reaching the minimum BOR value (~1.75 V), the reset state is released after 1 ms.

6 Suggestion

6.1 Printed Circuit Board

Due to technical reasons, it is best to use a separate layer of a multi-layer PCB for grounding (VSS) and another layer for VDD power supply. This can achieve good decoupling and shielding effects. For many applications, such boards cannot be used due to economic requirements. In this case, the most important feature is to ensure that VSS and the power supply have a good structure.

6.2 Component position

The initial layout of the PCB must distinguish the electromagnetic interference (EMI) contributions of different circuits. This reduces cross-coupling on the PCB, such as noise, high-current circuits, low-voltage circuits, and digital sections.

6.3 Grounding and power supply (VSS, VDD)

The VSS should be grounded at a single point in each module (noise, low-level sensitive, and digital) respectively to collect all the ground return currents. Loops must be avoided or made to have the smallest possible area. Power supply should be placed close to the ground line to minimize the area of the power supply loop. This is because the power supply loop behaves like an antenna and is thus a major emitter and receiver of EMI. All unoccupied PCB surfaces must be filled with additional ground to create shielding (especially when using single-layer PCBs).

6.4 Decoupling

The standard decoupling capacitor for an external power supply is a 1 μF ceramic capacitor. To reduce the area of the current loop, the supplementary 100 nF capacitor must be placed as close as possible to the VSS/VDD macro pins.

In general, decoupling all sensitive or noise signals improves electromagnetic compatibility (EMC) performance.

There are two types of decouplers:

Capacitors close to components. The inductance characteristics of all capacitors at frequencies outside a certain range must be taken into account. If possible, gradually decreasing capacitors (0.1, 0.01, ... μF) should be connected in parallel.

· Inductors. For instance, although ferrite beads are often overlooked, they are good inductors because they have excellent EMI energy distribution characteristics and no DC voltage drop (unlike when using a simple resistor).

6.5 Other signals

When designing an application, the following areas need to be studied in detail to improve EMC performance:

- Noise signal (clock)

Sensitive signals (high impedance)

Temporary interference can permanently affect the signals used for the running process, such as interrupt and handshake strobe signals (but not LED instructions).

For these signals, using a ground track around the vss can improve EMC performance, just as shorter lengths or noise-free and sensitive tracks (crosstalk influence) do.

For digital signals, the two logic states must reach the best possible electrical boundaries. It is recommended to use a slow Schmitt trigger to eliminate parasitic states.

6.6 Unused I/O and Features

Microcontrollers are designed for a variety of applications, and usually an application does not utilize 100% of the microcontroller's resources.

To avoid unnecessary power consumption (especially important for battery-powered applications) and improve EMC performance, idle clocks, counters or I/O should not be left floating. I/O should be externally pulled up or down to idle I/O pins, and unused functions should be "frozen" or disabled.

Alternatively, unused I/Os can be programmed as push-pull "low" to keep them at a specified level without using external components. However, in this case, the I/Os are not driven until the configuration is completed during power-on, which will result in a small amount of additional power consumption and may not be suitable for applications that are highly sensitive to power consumption.

6.7 User options

The STM8L and STM8AL devices feature user option characteristics that can be used to remap or enable/disable an automatic reset or low-speed watchdog. For details, please refer to the product data sheet.

6.8 Bootstrap program

The STM8L05xx, STM8L15xxx, STM8L162xx, STM8AL31xx and STM8AL3Lxx devices integrate a bootloader in ROM memory. Through this firmware, the device memory can be reprogrammed via the following interfaces:

USART communication interface for medium-capacity devices

- USART1, USART2, USART3, SPI1 and SPI2 communication interfaces for medium and above or high-capacity devices.

7 Reference Design

7.1 Component Reference

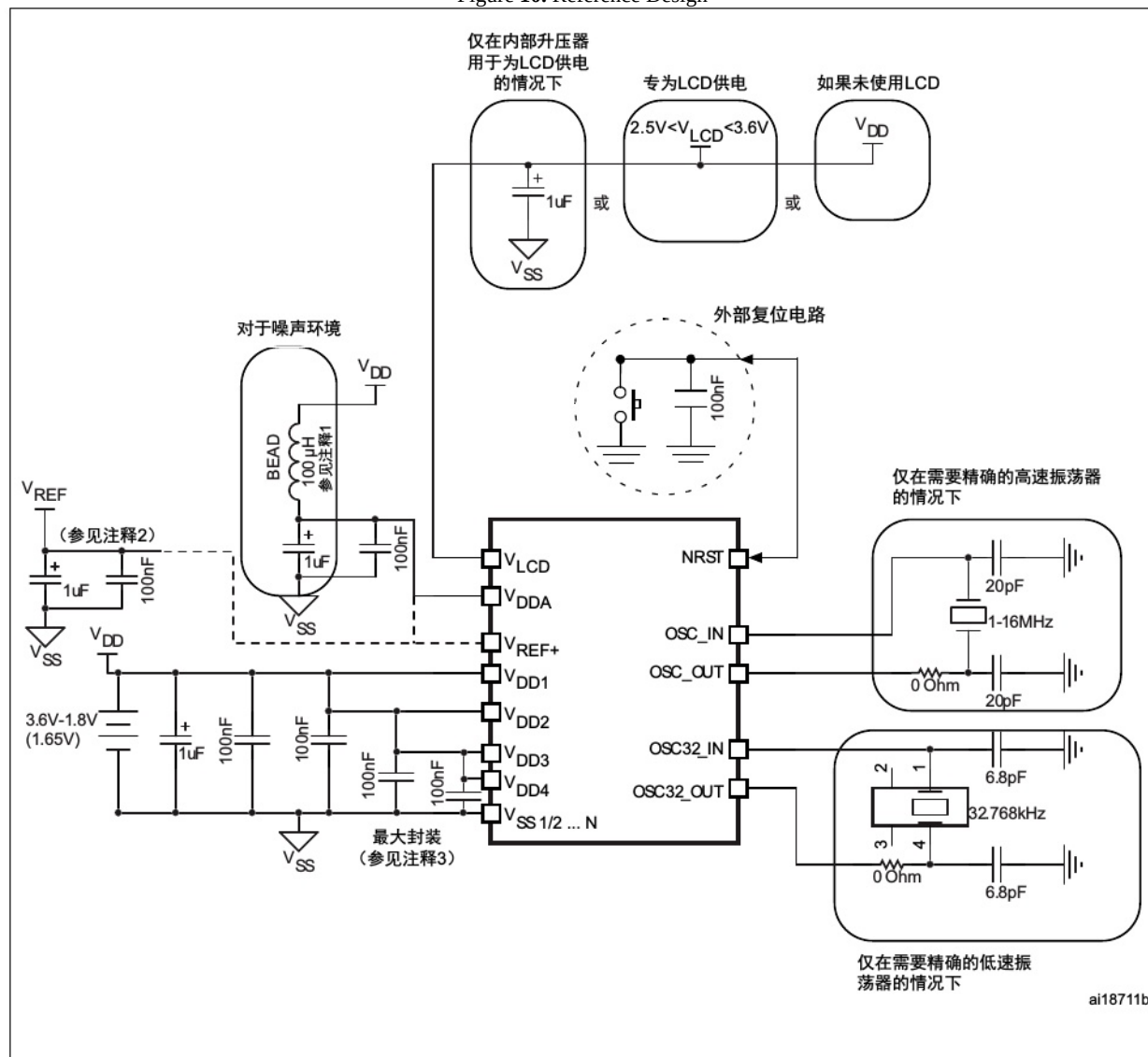
Table 2. Component List

ID	Component Name	Reference	quantity	Notes
1	microcontroller	STM8L, STM8AL	1	Refer to the "Pin Arrangement and Pin Description" and "Package Characteristics" sections of the STM8L and STM8AL data sheets to select the correct package.
2	battery	1.65 V to 3.6 V (1)	1	When BOR is enabled, the minimum value is 1.8 V.
3	capacitance	1 μ F	n	Decoupling capacitor
4	capacitance	100 nF	n	Ceramic capacitors (decoupling capacitors)
The following components are optional.				
5	capacitance	10 μ F	1	Ceramic capacitors (decoupling capacitors)
6	crystal oscillator	1 to 16 MHz	1	-
7	capacitance	20 to 40 pF	2	For crystal oscillator
	crystal oscillator	32 kHz	1	-
8	capacitance	5 to 20 pF	2	For crystal oscillator
	Ferrite magnetic ring			Depending on the noise being filtered
9	SWIM connector	Four pins	1	-

1.8 V to 3.6 V (for STM8L05xxx).

7.2 Schematic diagram

Figure 10. Reference Design



If these components are removed, short connections must be used to replace them.

2. Optional: If a separate external reference voltage is applied to VREF+, a 100 nF and a 1 μF capacitor must be connected to this pin. VREF+ is connected to VDDA or VREF.

For each VDDx pin, use a 100 nF ceramic capacitor and a separate 1 μF tantalum capacitor or ceramic capacitor.

8 STM8 Development Tools

The following tools are usually required:

- STVD is used for integrated development environments
- STM8 C compiler (from Cosmic, Raisonance or IAR)
- ST tool suite and STM8 firmware libraries from STMicroelectronics, including the STM8L101xx Standard Peripheral Library, and the STM8L05x/STM8L15x/STM8L16x/STM8AL31x/STM8AL3Lx Standard Peripheral Libraries.
- STMicroelectronics' STM8 evaluation boards (STM8L101-EVAL for STM8L101xx, STM8L1526-EVAL for medium-capacity STM8L15xxx and STM8AL31xx/STM8AL3Lxx, and STM8L1528-EVAL for high-capacity STM8L15xxx/STM8L162xx)

If the user uses the STM8L101-EVAL, they also need the "Rlink" from Raisonance and the HW SWIM debugging interface of ST-Link or STice-SWIM. The debugging interface ST-LINK is included in the STM8L1526-EVAL and STM8L1528-EVAL.

- STM8L-DISCOVERY

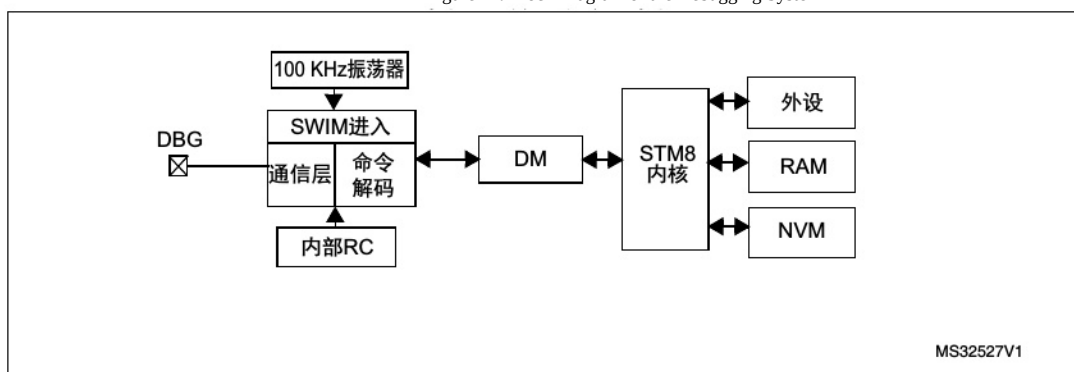
8.1 Single Wire Interface Module (SWIM)

8.1.1 SWIM 概述 8.1.1 SWIM Overview

The online debugging mode or online programming mode is managed through a single-line hardware interface, which is based on an open-drain line and features ultra-fast memory programming capabilities. Besides being coupled with the online debugging module, SWIM can also perform non-intrusive read/write operations on RAM and peripherals. This makes the online debugger very powerful, approaching the performance of a full-featured emulator.

The SWIM pin can be used as a standard I/O (with an 8 mA capability). However, if the user wants to use it for debugging, there are some limitations. The safest way is to use it as a jumper selection on the PCB. For more information on the SWIM protocol, please refer to the STM8 SWIM Communication Protocol and Debugging Module User Manual (UM0470).

Figure 11. Block Diagram of the Debugging System



8.1.2 SWIM connector pins

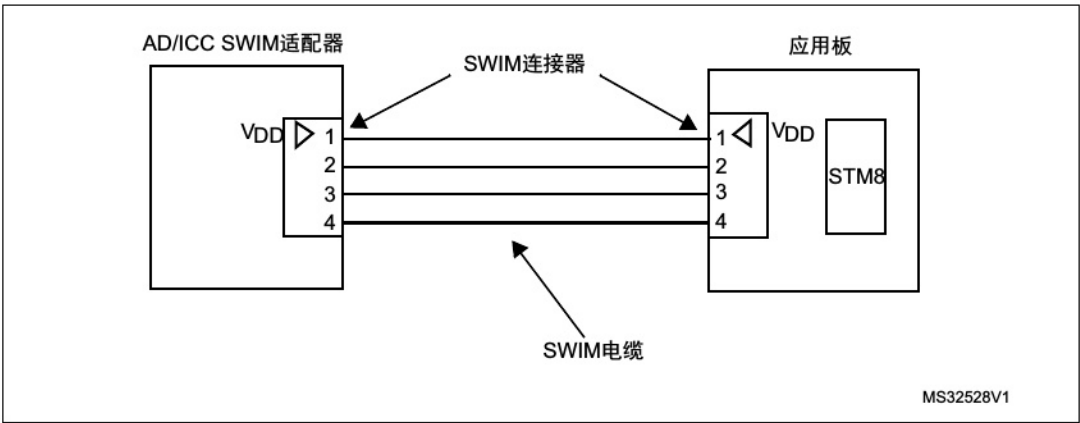
The SWIM connector pins include the four pins described in [Table 3](#).

Table 3. SWIM Connector Pins

Pin number	Pin name
Pin 1	VDD
Pin 2	SWIM pin
Pin 3	Vss
Pin 4	Reset

8.1.3 Hardware connection

Figure 12. Hardware Connection



Note: It is recommended to place the SWIM header as close as possible to the STM8L/STM8AL device, as this will reduce any possible interference caused by long leads.
Signal degradation caused by PCB traces.

8.2 STice Simulator

8.2.1 STice Overview

STice is a modular high-end simulation system that connects the PC and the application board via a USB interface, rather than the target microcontroller.

It is supported by the free STM8 tool suite: IDE ST Visual Develop (STVD), programmer ST Visual Programmer (STVP), and STM8 assembler. For more detailed information, please refer to the STice simulator for STM8.

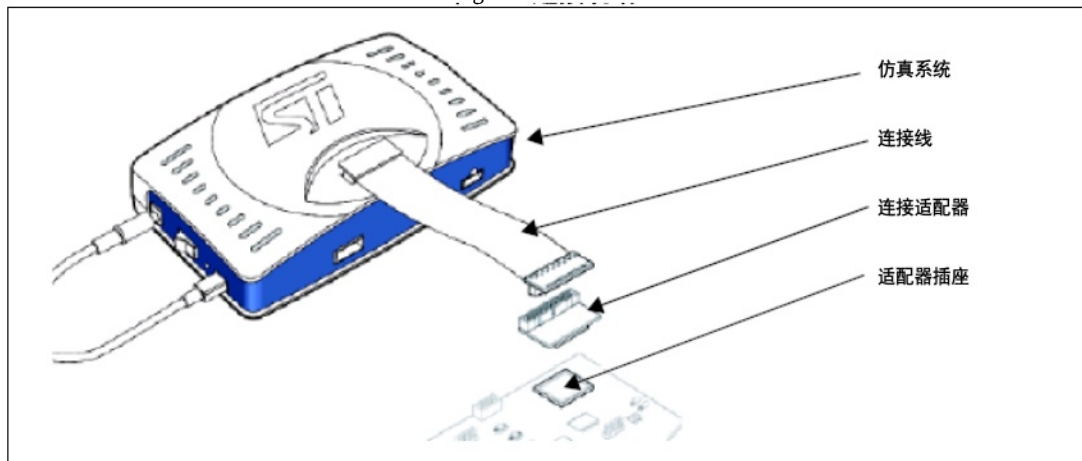
STice has two different working modes. This section will introduce both of them:

Simulation mode

Online mode

It can also replace RLink for SWIM connections.

Figure 13. Connection Instructions



Simulation System: STice

Simulation box

Cables for USB, power, trigger and analyzer input

Connecting wire

60-pin or 120-pin cables for connecting application boards

Connection adapter

Connect the connection wires to the STM8L/STM8AL microcontroller package.

Adapter socket

- Specific socket for connecting the adapter to the STM8L/STM8AL microcontroller.

8.2.2 STice Simulation Configuration

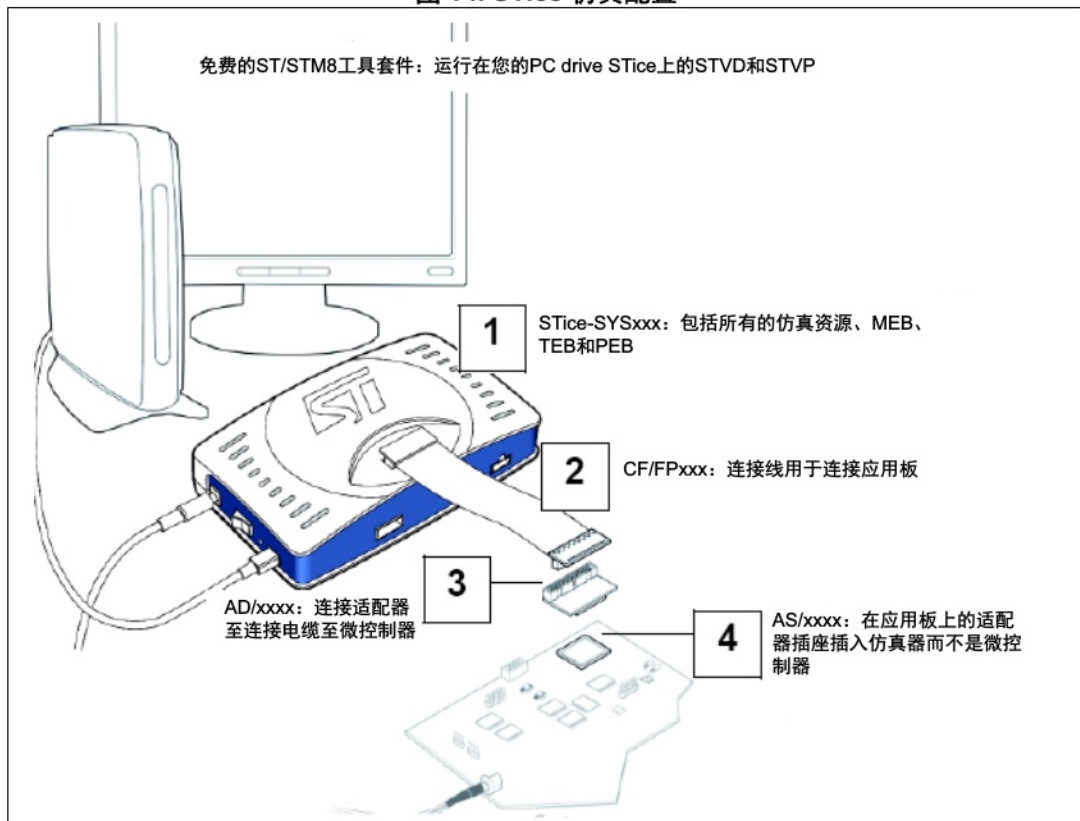
In the simulation configuration, STice connects the PC and the application board via the USB interface instead of using the target microcontroller.

- Connection cable: Flexible cable (60-pin or 120-pin, depending on the target microcontroller), used to transfer signals from the STice to the application board.
- Connection adapter: Connect the connection line to the target microcontroller package on the user application board.

Adapter socket: A socket welded to the application board rather than the microcontroller, which simultaneously receives the connection adapter.

The STice system does not include the above-mentioned accessories. To determine exactly what is needed for the supported microcontrollers, refer to the online product selector on the www.st.com website.

图 14. STice 仿真配置

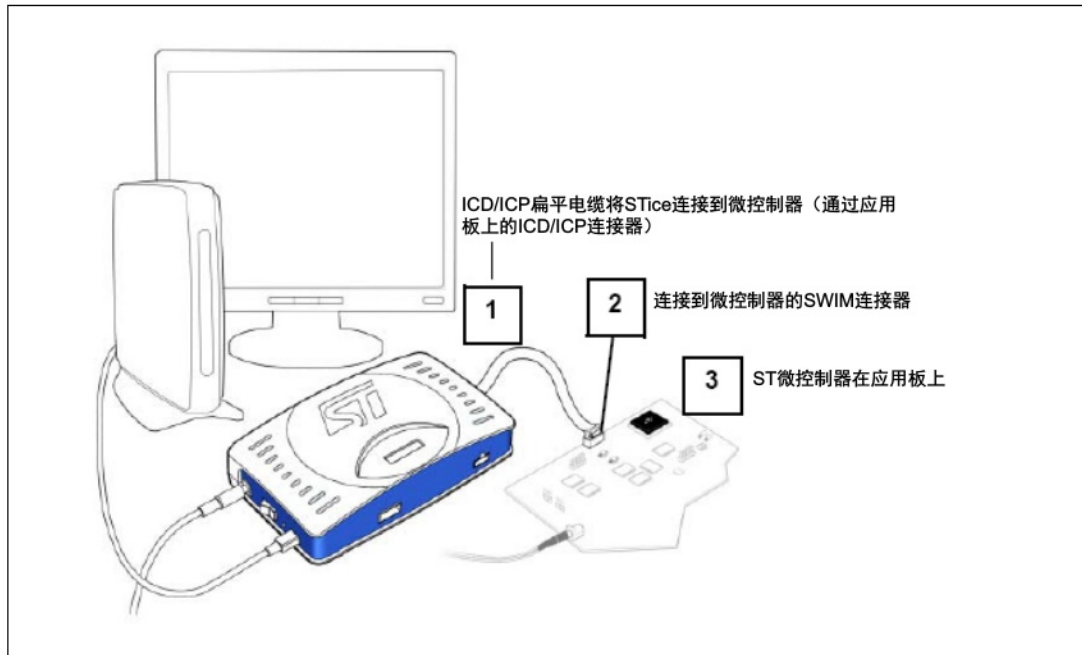


8.2.3 Online programming and debugging

During the online debugging/programming configuration process, when the application is running on the microcontroller of the application board, STIcice enables programming and debugging of the application on the microcontroller. STIcice supports the SWIM protocol, which allows for online programming and debugging of the microcontroller using only one general-purpose I/O.

In simulation and online programming/debugging configurations, STIcice is driven by the ST Visual Develop (STVD) or ST Visual Programmer (STVP) integrated development environment running on the main PC. This provides a single, easy-to-use interface that offers full control over advanced application building, debugging, and programming features.

Figure 15. Online Programming and Debugging



8.3 RLink and STLink

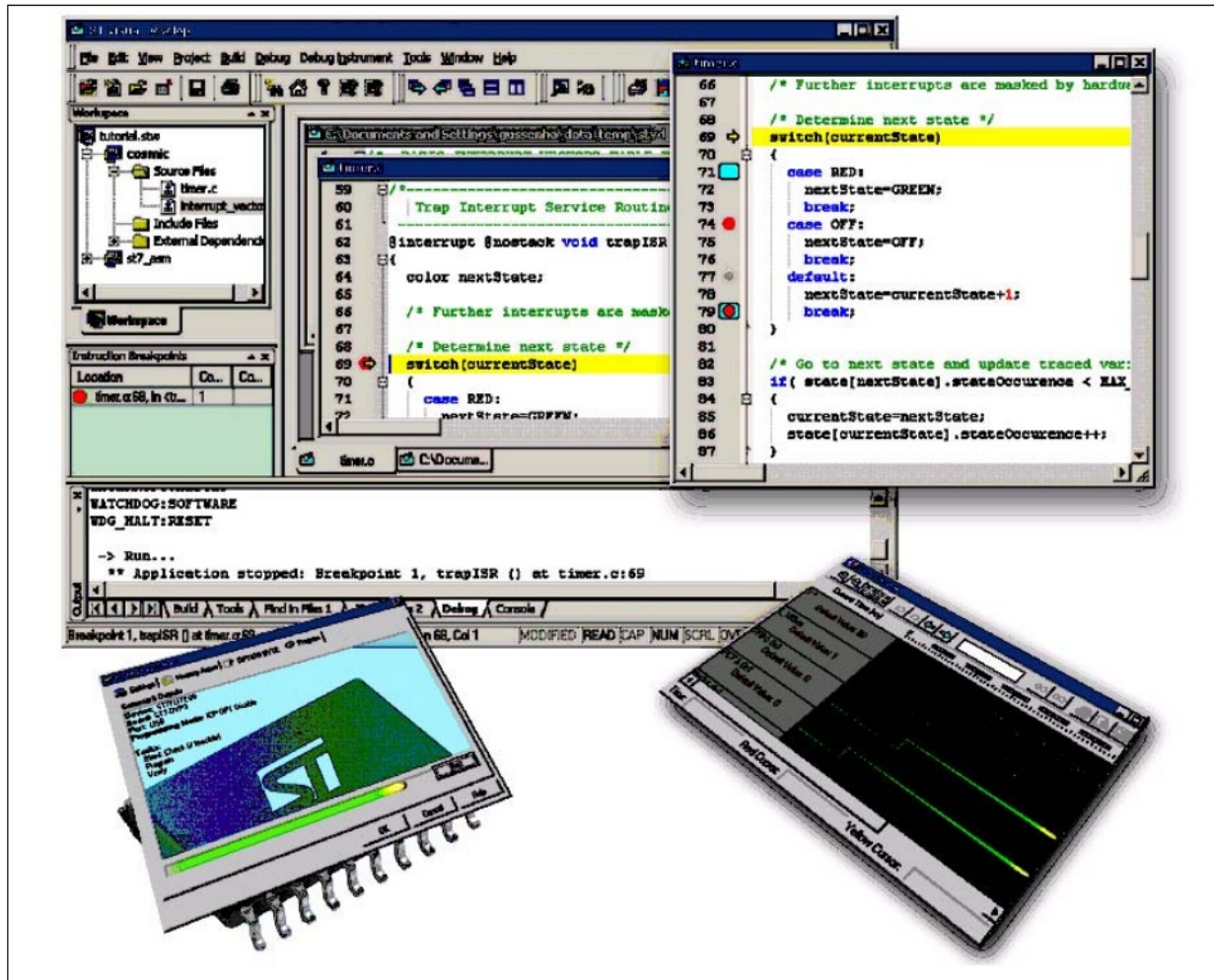
The RLink and STLink debugging tools can connect the STM8L evaluation board or any user application board with a SWIM interface to a host PC via USB for programming and debugging. Please refer to [Section 10.3.3: Connecting Hardware](#) on [page 34](#).

9 STM8 Software Toolchain

To write, compile and run the first software on STM8L/STM8AL devices, the following components of the software toolchain are required (see [Figure 16](#)):

- Integrated Development Environment
- compiler
 - Firmware library (optional, for convenient startup)

Figure 16. STM8 Software Toolchain



9.1 Integrated Development Environment

The integrated development environment ST Visual Develop (STVD) provides a user-friendly and efficient environment for the entire control application development process, from building and debugging application code to microcontroller programming. STVD is part of the free ST tool suite, which also includes the ST Visual Programmer (STVP) programming interface and ST Assembler Linker.

To build applications, STVD seamlessly integrates C and assembly language toolchains for ST, including Cosmic and Raisonance C language compilers and ST Assembler Linker. During debugging, STVD offers an integrated simulator (software) and supports a full range of hardware tools, including the low-cost RLink online debugger/programmer, the high-end STice simulator, and the low-cost ST-LINK tool.

To program applications for STM8L/STM8AL, STVD also provides an interface for reading from, writing to, and verifying the microcontroller's memory. This interface is based on ST Visual Programmer (STVP) and is suitable for all target devices and programming tools supported by STVP.

The free ST tool suite for STM8 can be obtained from the STMicroelectronics homepage (see www.st.com).

9.2 Compiler

STM8L/STM8AL devices can be programmed with a free assembly toolchain included in the ST tool suite.

Since the kernel is optimized for supporting high-level languages, it is recommended to use a C compiler!

The C compiler for STM8 is provided by third-party companies Cosmic, Raisonance and IAR.

A free version of a C compiler capable of generating up to 32 Kbytes of code can be obtained from www.cosmic-software.com and www.raisonance.com.

9.3 Firmware Library

For each STM8 peripheral, the STM8 firmware library is a complete set of source code examples. It is written in strict ANSI-C and is fully compatible with MISRA C 2004.

All the examples can be used with four workspaces and project definition files: one for STVD and Cosmic C compiler, one for STVD and Raisonance compiler, one for Raisonance Integrated Development Environment and compiler (RIDE7 IDE), and one for IAR Embedded Workbench for STM8 (EWSTM8). This enables users to easily load and compile in their preferred development environment.

The example running on the STMicroelectronics STM8L evaluation board can be easily ported to other types of hardware.

For more information and downloads of the STM8L/STM8AL firmware library, please visit www.st.com/mcu.

10 Set up the STM8 development environment

10.1 Install Tools

All software tools come with a setup wizard to guide users throughout the installation process. It is recommended to install the tools in the following order:

C compiler

2. ST Tool Suite

3. STM8 Firmware Library

ST-LINK does not require the installation of any dedicated software in the STM8 development environment because the ST tool suite comes with the necessary drivers.

The R-link drive must be started separately from the following location:

Start/Programs/STtoolset/Setup/Install Rlink driver.

10.2 Tools Used

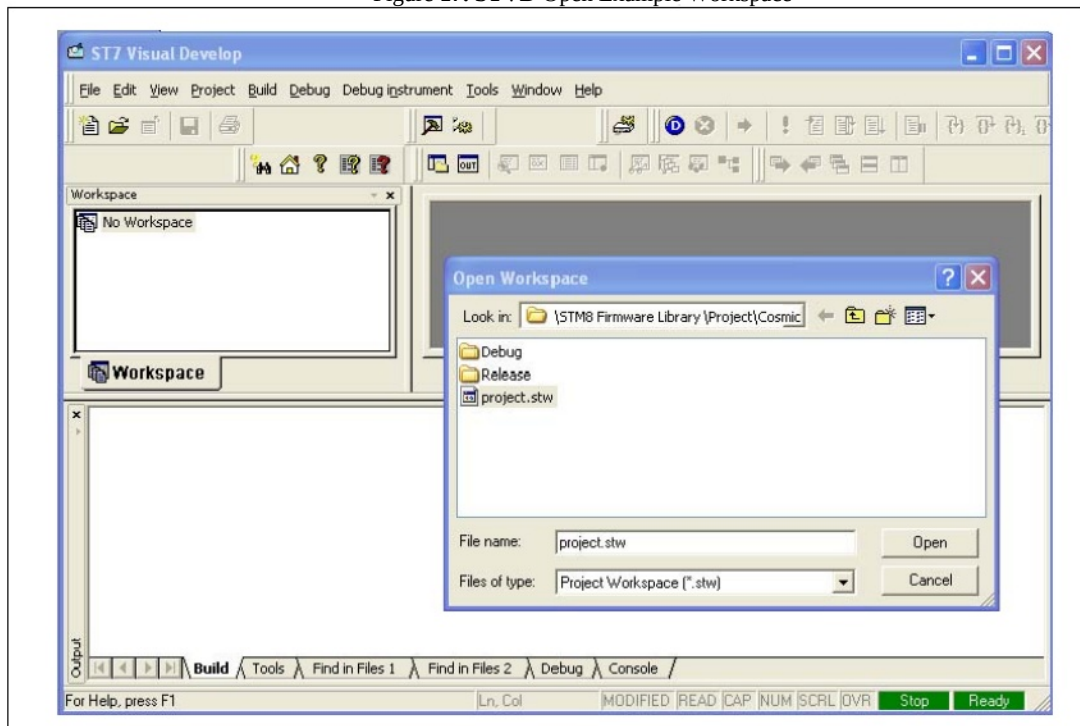
Once the tool is installed, the ST Visual Develop (STVD) integrated development environment can be launched.

Users can choose to create a new workspace for a new project or open an existing workspace. If it is the first time using STVD, it is recommended to open an existing project from the STM8 firmware library.

Note: Even if the user does not plan to use the library, the existing library project can be used as a template for configuring all compiler options. Enter the code after *main()*.

The STM8 firmware library includes several examples for each peripheral and a workspace, which contains a blank project ready to accept user C code. It is located in the subdirectory \Project\Template of the firmware (see Figure 17). Users can choose between STVD\Cosmic, STVD\Raisonance, RIDE or EWSTM8.

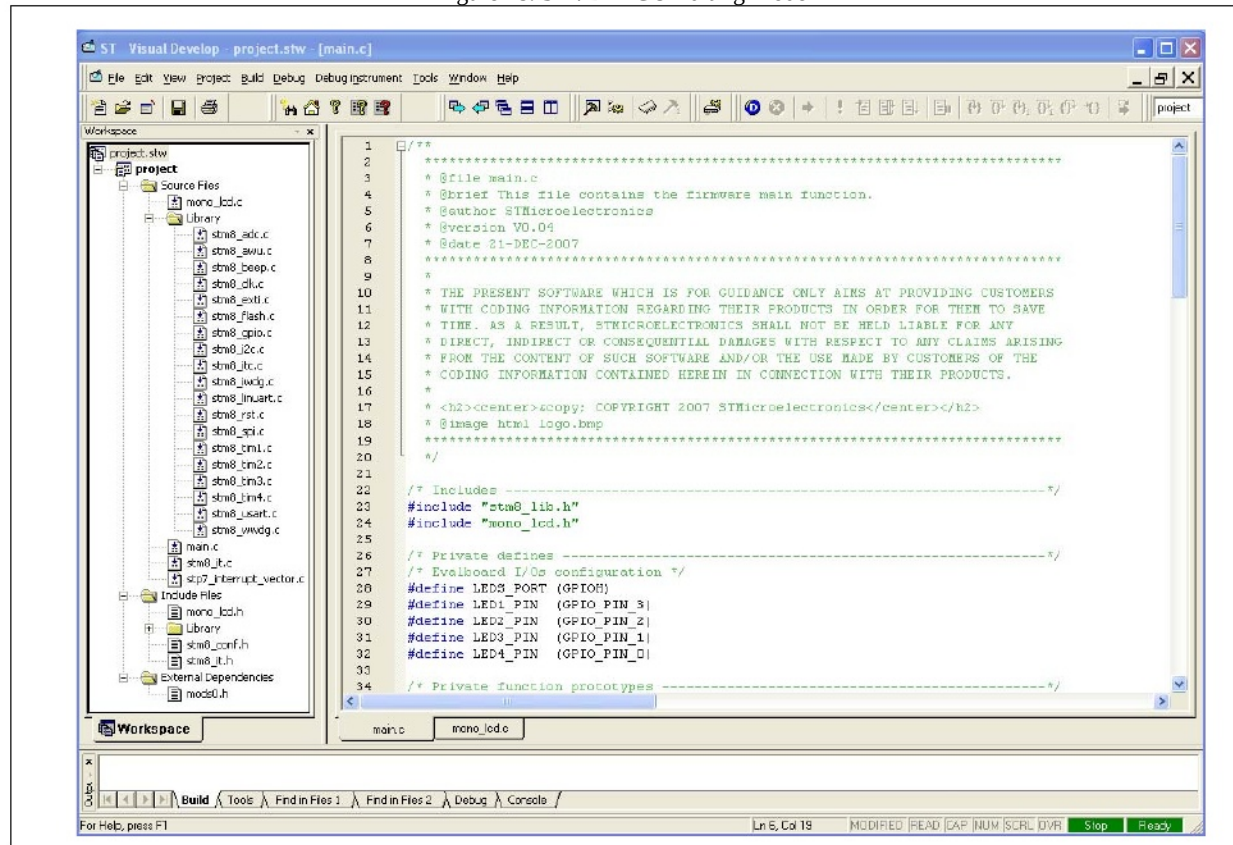
Figure 17. STVD Open Example Workspace



10.2.1 Project Editing

All project source files are visible and editable (see [Figure 18](#)).

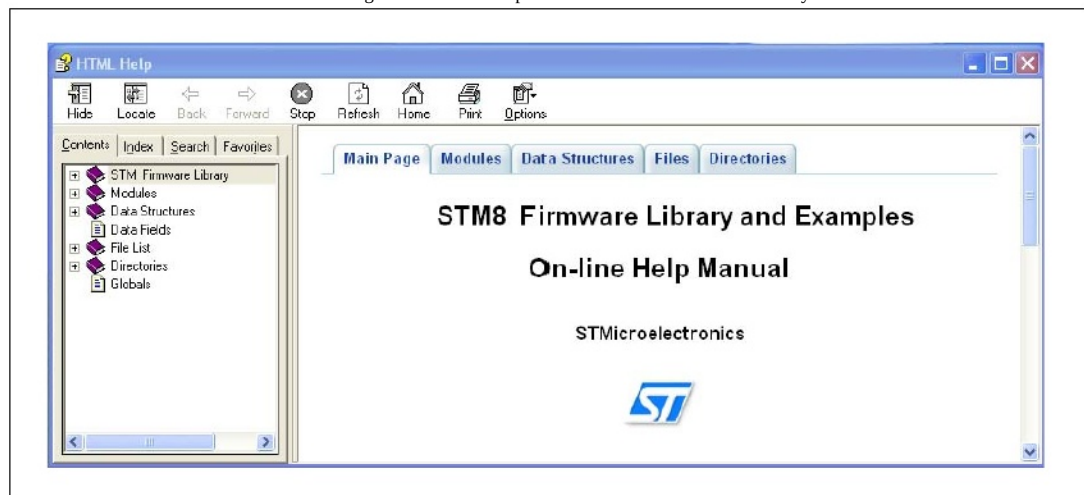
Figure 18. STVD MCU Editing Mode



10.2.2 Online Help

The online help manual can be obtained from the firmware installation directory (see [Figure 19](#)), and it is used to help users understand the structure of the STM8 firmware library.

Figure 19. Online Help Manual for STM8 Firmware Library



10.3 Run the demonstration software

Visit www.st.com/mcu to find STM8L/STM8AL products.

- Select the firmware for STM8L1x-EVAL, STM8L1526-EVAL or STM8L1528-EVAL.

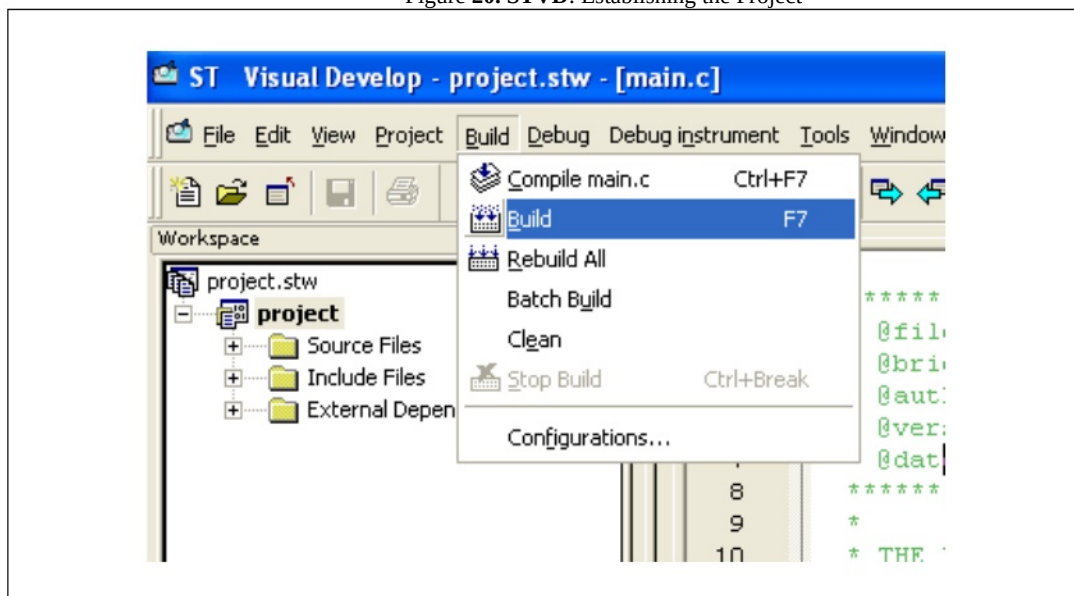
Open the required project workspace in the selected demo firmware package.

To run the demonstration software on the STM8 evaluation board, the project needs to be compiled, and the correct HW tool must be selected before starting the debugging session.

10.3.1 Compile the Project

The project can be compiled using the "Build" function in the "Build" menu (see [Figure 20](#)).

Figure 20. STVD: Establishing the Project

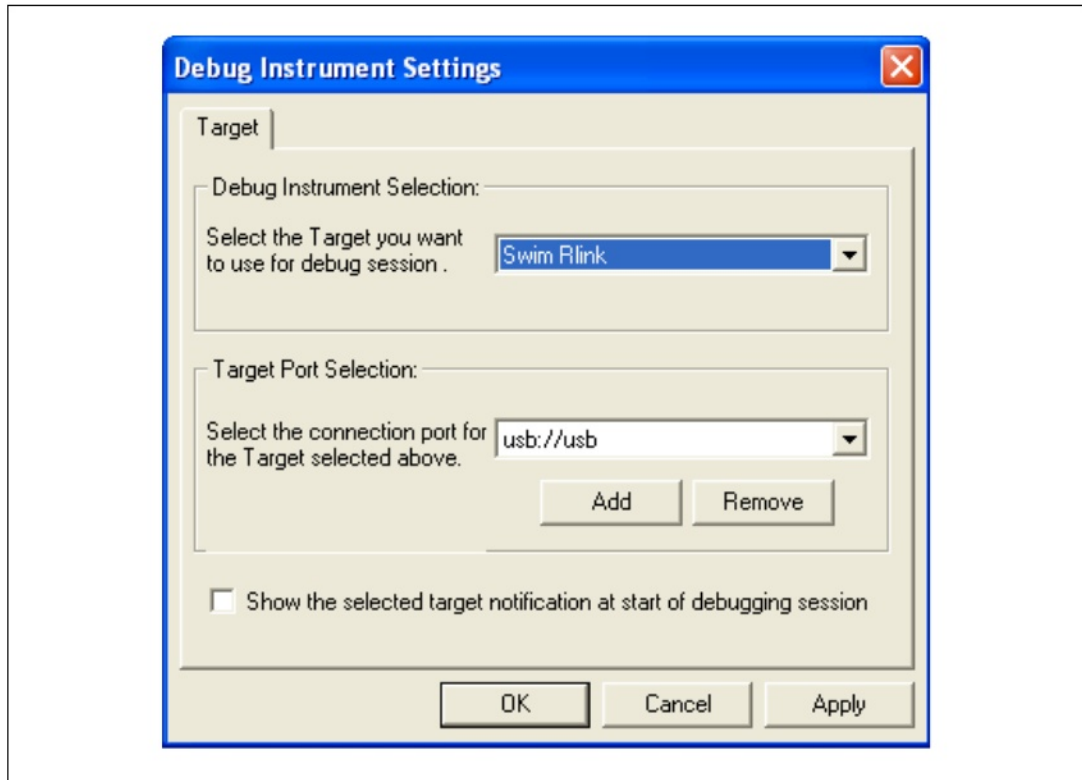


10.3.2 Select the Correct Debugging Tools

In the following example, the Rlink tool is used to communicate with the on-board debug module of the STM8 via the SWIM interface.

The Rlink tool can be selected from the "Debug Instrument Selection" in the "Debug Instrument Settings" dialog box (see [Figure 21](#)).

Figure 21. STVD: Select Debugging Tools



10.3.3 Connecting Hardware

The debugging tool STLink is integrated on the STM8L1526-EVAL and STM8L1528-EVAL boards. Users can connect a PC to the USB connector. This connection ensures both debugging connection and power supply. If the jumper on the board is not in the default position, please refer to the evaluation board user manual to select the power supply and debugging support jumpers.

For the STM8L101-EVAL, the Rlink tool can be connected to a PC via a standard USB port and is also powered through this USB interface. On the controller side, the connection to the STM8 evaluation board is made using a SWIM interface cable. The STM8L101-EVAL evaluation board is powered by an external 5 V power supply (see [Figure 22](#)).

Figure 22. Connecting the debugging tool to the STM8L101-EVAL evaluation board



Figure 23. Connecting the debugging tool to the STM8L152x-EVAL evaluation board

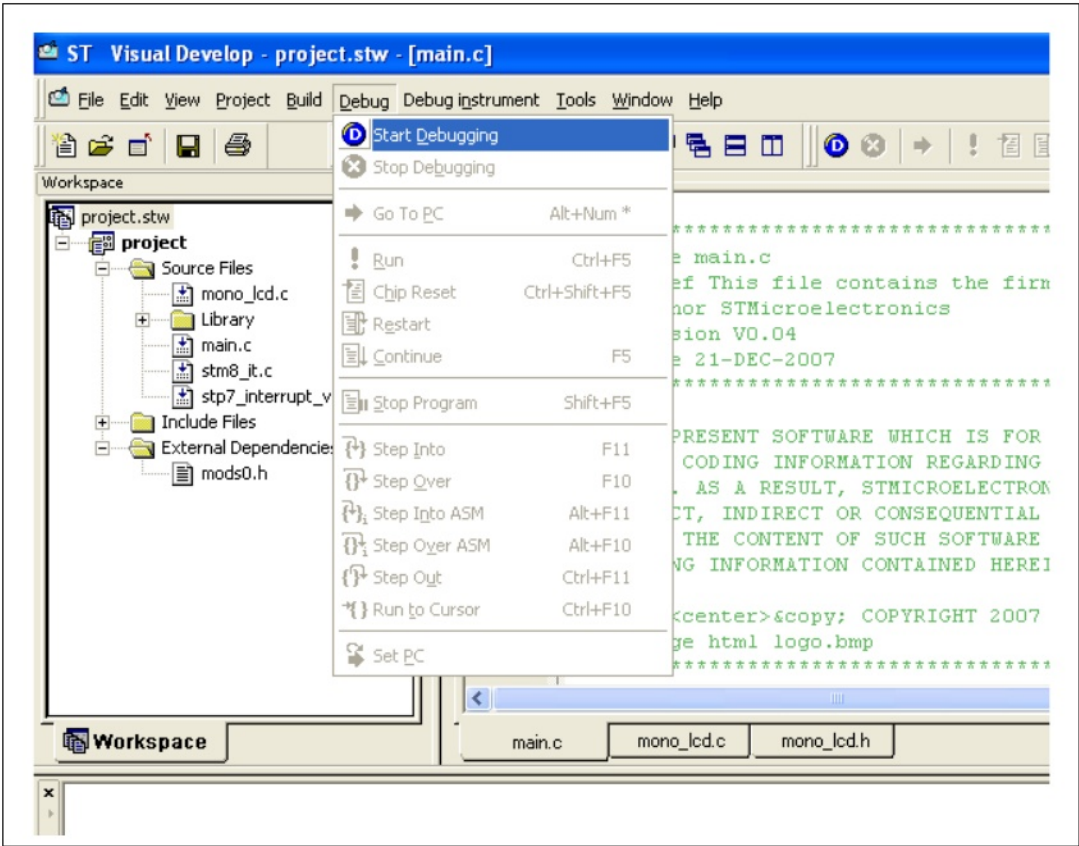


Note: On the STM8 Rlink adapter board, the "SWIM" jumper must be set. If the SWIM line is not pulled up in the application, Then, the "ADAPT" jumper must be set. Do not set the "PW-5V" and "12MHz" jumpers.

10.3.4 Start Debugging Session

The debug mode can be entered by using the "Debug Start Debugging" command (see Figure 24).

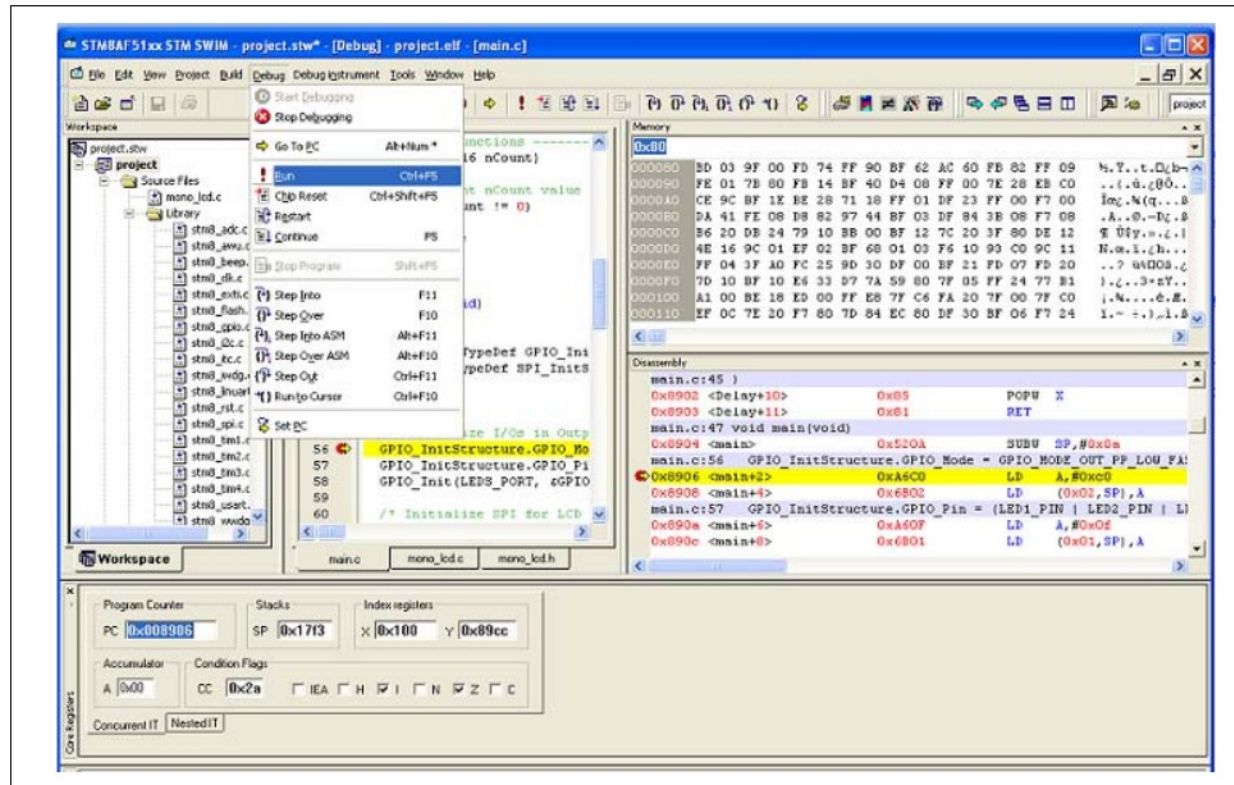
Figure 24. STVD: Start Debugging Session



10.3.5 Run the software

After entering the debug mode, the software can be started through the "Debug Run" menu command (see Figure 25).

Figure 25. STVD: Running Software



The LCD display on the STM8 evaluation board indicates that the debugging session has been successfully completed (see [Figure 26](#)).

Figure 26. STM8 Evaluation Board



10.3.6 Subsequent Operations

Then, starting from the initial debugging session described above, the additional peripherals of the STM8L/STM8AL devices can be enabled and run one by one.

Many features of the STM8L/STM8AL devices are supported by dedicated hardware on the STM8 evaluation board. The STM8L1x firmware library provides the necessary software drivers, including STM8L/STM8AL peripheral drivers (USART, ADC, SPI) and drivers for the EVAL board modules (LCD, serial memory).

11 Documentation and Online Help

Documentation resources related to tool usage include:

Application

- STM8L/STM8AL Data Sheet

How to Program the Flash Program Memory and Data EEPROM of STM8L (PM0054)

STM8 CPU Programming Manual (PM0044)

STM8L05xx, STM8L15xx, STM8L162x, STM8AL31xx and STM8AL3Lxx microcontroller series (RM0031)

- STM8L101xx Microcontroller Series (RM0013)

tool

- STM8L/STM8AL firmware library and version description (included as a help file with detailed firmware library description)

STice advanced simulation system for ST microcontroller data briefs

- STice User Manual

User manuals for Cosmic, Raisonance or IAR C compilers

User manual for STM8L101-EVAL, STM8L1526-EVAL or STM8L1528-EVAL evaluation boards.

- STM8L1x-EVAL, STM8L1526-EVAL or STM8L1528-EVAL firmware

- ST Visual Develop Guide (as a help file in the ST toolchain)

User Manual for ST Visual Develop (STVD)

STM8 SWIM Communication Protocol and Debug Module User Manual (UM0470)

Developers can exchange ideas in the microcontroller discussion forum on www.st.com. This is the best place to find different application ideas. In addition, the website also has a FAQ technical resource about microcontrollers, which provides many questions and solutions.

12 Revision History

Table 4. Document Revision History

Date	Version	Change
September 9, 2009	1	Initial version
16-Sep-2010	2	For medium-capacity, medium-plus and high-capacity devices, all chapters and references have been updated. The STM8L16x devices have been added. VDDIO and VSSIO were respectively replaced by VDDx and VSSx. Some 48-pin packages were replaced with certain encapsulations. The IAR C compiler has been added. Section 2.1: Power Supply Overview: Text updated. Figure 1: Added annotations. 2: Remove the + symbol on VLCD. Crystal oscillator / Ceramic resonator: Reference to AN2867 has been added. Section 3.2: Analog Input: Text updated. Section 4.3.2: LSE Clock: Reorganize the chapter. Section 5.1: Overview of Reset Management: Removed "Electromagnetic Sensitivity (EMS) Reset (present only in STM8S series)." Figure 7: Removed EMS reset; added external reset circuit. Section 5.2: Hardware Reset Implementation: Text about pull-down capacitors has been added. Table 2: Component List: Updated the "Comment" information for ID No. 2 (Battery); Added ID No. 5 (Ceramic Capacitor). Figure 10: Added external reset circuit and annotation 1. Section 9.2: Compiler: Replaced "16 Kbytes" with "32 Kbytes" in the generated code. Section 9.3: Firmware Library: Updated the text explaining the workspace and project definition files. Section 10: Setting up the STM8 Development Environment: The introduction has been removed. Section 10.2: Using Tools: Added EWSTM8.
March 7, 2011	3	Figure 1: Updated image content and footnote. Section 2.1: Power Supply Overview: Updated content for each item. Section 3.1: Analog Circuit Power Supply: Updated content related to VREF+. Section 6.4: Decoupling: Updated the value of the bypass capacitor (1 μF instead of 100 μF). Figure 10: Updated image content and footnote. Section 8: STM8 Development Tools: Updated point 5 regarding the STM8L101-EVAL. Section 9.1: Integrated Development Environment: Added the low-cost ST-Link tool to the list of supported hardware tools for STVD.
November 6, 2012	4	Update the document to include the STM8AL products. Add Table 1: Applicable Products. Add note (1) to Table 2: Component List. Update the references in Section 8: STM8 Development Tools.
July 23, 2013	5	Updated: - Section 2.1: Power Supply Overview - Section 2.3: Power-On/Power-Down Reset (POR/PDR)

Table 5. Revision History of the Chinese Document

Date	Version	Change
October 12, 2015	1	Initial Chinese version

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